

# Charles Emmanuel A. Cruz

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## Data Science Projects & Hackathons

### E-commerce Fraudulent Product Detection | Top 9 Teams – Shopee Tr[AI]blaze Hackathon

July 2026

- Developed a fraudulent product detection system utilizing RoBERTa for text description analysis and Isolation Forest for detecting pricing anomalies, combining outputs into a unified product legitimacy trust score.
- Fine-tuned transformer models using a dataset of 40,000 product records, achieving a classification precision of ~89%.
- Integrated a human-in-the-loop (HITL) review system for low-confidence classifications, improving overall model reliability and categorization accuracy.

### Cross-platform E-commerce Store Identity Verifier | Top 9 Teams – Shopee Tr[AI]blaze Hackathon

July 2026

- Designed and implemented a programmatic cascading pipeline to handle common identity verification cases, significantly reducing reliance on expensive LLM API calls.
- Reduced token consumption by a baseline of 1.05 billion tokens per 1 million records, resulting in estimated cost savings of over \$3,000 per million records evaluated.
- Engineered a human-in-the-loop dashboard to review, edit, and archive existing LLM rules based on production usage metrics.

### An IoT-Based Bus Tracking and Passenger Monitoring System for Jasper Jean Services Inc.

April 2026

- Developed an IoT-based bus tracking system integrating GPS hardware and MQTT protocols for real-time telemetry data transmission.
- Implemented a YOLO-based (Ultralytics) computer vision model to perform real-time passenger counting and crowd density analysis prior to dashboard transmission.
- Architected a custom web dashboard integrating REST APIs to synchronize and visualize real-time spatial and passenger data.

## Professional Experience

### Department of Science and Technology – Planning and Evaluation Service

Taguig City

#### Backend Developer Intern

January – April 2026

- Engineered RESTful APIs to support backend services, optimizing data exchange protocols and enhancing system interoperability across a microservices architecture.
- Implemented Apache Kafka for asynchronous message queueing, establishing reliable event-driven data synchronization and data pipelines between core operational services.
- Collaborated in an Agile development environment, actively participating in sprint planning and code reviews to align technical backend solutions with organizational objectives.

## Education

### De La Salle University – Dasmariñas

Dasmariñas, Cavite

BS Information Technology (GPA: 3.86/4.00)

Expected August 2026

**Awards:** Summa Cum Laude, DOST-SEI Merit Scholar

**Competitions:** Provincial Champion – Philippine Statistics Quiz, Participant – UP Diliman Stat-is-eeks

## Technical Skills

**Machine Learning & AI:** Scikit-Learn, PyTorch, Hugging Face Transformers, Natural Language Processing (NLP), Computer Vision, Ensemble Methods

**Data Science Libraries:** Pandas, NumPy, Matplotlib, Seaborn, SciPy

**Programming Languages:** Python, SQL, JavaScript, C#, C++

**Statistical Analysis:** Hypothesis Testing, A/B Testing, Probability Distributions, Correlation & Regression Analysis

**BI Tools & Platforms:** Power BI, Tableau, Google Looker, Excel, SAP S/4HANA, Odoo, Jupyter Notebook, Google CoLab

## Certifications & Certificates

- IT Specialist – Databases (Certiport)
- PCAP: Programming Essentials in Python (OpenEDG)
- Intermediate Python | Understanding Data Engineering | Data Manipulation with pandas (DataCamp)